
EXECUTIVE SUMMARY

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AI infrastructure orchestration for GPU cost optimization

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Overview

Aurelius is an infrastructure orchestration layer that reduces GPU operating cost through SLA-aware workload placement, time-shifting, and market-aware scheduling. It decides when and where flexible compute runs so the same work lands in lower-cost hours and regions, without changing the work itself.

Aurelius is a decision layer, not an execution platform. It produces placement and timing decisions that an existing scheduler carries out. It does not take custody of workloads, requires no privileged access to begin, and operates in a read-only posture by default.

The opportunity

GPU compute is the dominant variable cost in AI infrastructure, and the energy that powers it is priced in markets that move hour to hour and differ region to region. Wholesale electricity prices routinely vary by a factor of two or more across a day and across interconnects, yet most schedulers place jobs as they arrive, in a default region, regardless of when or where the work would be cheapest to run.

A large share of GPU work — training, fine-tuning, batch inference, data processing, maintenance — carries real scheduling slack that goes unused. The result is a recurring, structural overspend that is invisible on any single invoice and difficult to address manually at fleet scale.

How it works

For each schedulable job, Aurelius forecasts near-term wholesale price movements across the relevant markets and selects the region and start window that minimizes energy cost, subject to the job's deadline, SLA class, and the capacity of each region. Savings derive from three mechanisms, in order of contribution: time-shifting flexible jobs into lower-priced hours; routing across regional price spreads; and rescheduling eligible long-running jobs when a materially better price emerges. Because all three depend on flexibility, savings scale with the share of the fleet that can tolerate delay or relocation.

FIGURE

Market + workload signals

Ingestion » Forecasting » Optimization » Safety gate
Decision » Reporting (replay / shadow) or scoped execution

Decision pipeline. Aurelius owns the decision; the customer's scheduler owns execution.

Validation status

Tier 1 region and time placement has been validated in leakage-free historical replay on real day-ahead prices from CAISO, PJM, and ERCOT. Measured against a strong baseline that always routes to the cheapest region at submission using live prices, the observed mean cost reduction across seven workload classes was 25.0% on Q1 2026 data and 22.8% on Summer 2025 data. These are historical-replay observations, not guarantees; customer-specific savings are confirmed in shadow mode before any commitment.

Workload class	Observed reduction (p50)	Driver
Background maintenance	~40%	Fully flexible, freely reschedulable
Data processing	~38%	High flexibility, short duration
LLM batch inference	~34%	Batch, tolerant of ~24h delay
Scheduled batch	~25%	Moderate flexibility
Training	~15%	Long, often non-interruptible
Fine-tuning	~13%	Moderate; sensitive to volatility
Real-time inference	~10%	Cannot be delayed

Savings track flexibility. A fleet weighted toward batch and maintenance work sits at the upper end; a fleet weighted toward latency-hard inference sits at the lower end.

Why this is safe to adopt

Three design choices make Aurelius suitable for production adoption. Decisions are advisory and reversible: Aurelius emits placement and timing decisions and the existing scheduler executes them, so there is no custody of workloads. Behavior degrades to the status quo: when a forecast is missing or low-confidence, the decision falls back deterministically to the customer's current behavior, and the optimizer cannot produce an outcome worse than the baseline it is measured against. And savings are evidenced before they are claimed: shadow mode compares predicted savings to realized settlement prices on the customer's own workload before any decision is acted on.

Engagement model

A pilot proceeds through three non-invasive phases. The decision to move forward rests on the shadow-mode result measured on the customer's own footprint, not on the historical benchmark.

Phase	What it does	Customer exposure
Offline replay	Re-runs historical workload against historical prices	None — read-only analysis

Phase	What it does	Customer exposure
Shadow mode	Records live decisions; compares to realized prices	None — no workloads executed
Controlled execution	Acts on decisions for selected flexible workloads	Opt-in, policy-gated, reversible

Scope of what is validated

Validated today: Tier 1 region and time placement across the three largest U.S. wholesale markets.
Available but dependent on customer integration: cluster and queue-aware placement, GPU and node-level placement, broader carbon coverage, and European markets. These are described as roadmap, not as current capability, and none is a prerequisite for a first pilot.

Recommended next step

Provide a recent workload trace covering 30 to 90 days and confirm the compute regions and SLA classes involved. The first deliverable is an offline-replay projection, followed by a shadow-mode validation on the actual footprint. Both phases are read-only and require no change to production scheduling.